

Exploring Multiple-Display Interaction for Live XR Performance

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ABSTRACT

Although VR concerts offer a unique experience for audience to watch a performance, people still tend to participate in a live performance physically for the co-presence and shared experience are difficult to perceive in VR. To address this issue, we propose Actualities, a live XR performance system that integrates onsite and online concerts to create a seamless experience across multiple displays. Our system utilizes various sensors to detect signals from musical instruments and onsite audiences, digitalizing onsite performance elements into a virtual world. We project the visuals onto screens and live-stream the content for audiences to watch through various devices, and we also designed several interactive elements for the audience to interact with the public display. To evaluate our system, we conducted exploratory research to help us refine our system and improve the cross-reality experience.

CCS CONCEPTS

• Human-centered computing → Interaction design.

KEYWORDS

Multiple-Display Interaction, Extended Reality, Live Performance

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1 INTRODUCTION

Previous research has focused on increasing audience engagement and interaction to enhance immersion in VR concerts. For instance, LiVRation[Kasuya et al. 2019] enables users to watch the live performance from various perspectives and to control the volume of

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Figure 1: (a)Actualities, a seamless live XR performance system. (b)Audiences danced to create virtual animal avatars. (c)Audiences sent emoji feedback through their phone. (d)Viewpoint from the VR-HMD.

sound freely. [Horie et al. 2018] present a VR concert system which generates visual effects based on the user's biomedical signals, such as brainwaves and heartbeats. In these projects, the audience not only gains more autonomy, but even becomes a part of the performance. Meanwhile, presence is a crucial element in the VR concert experience. [Beacco et al. 2021] aim to create the illusion of presence, which makes users feel as though they are actually experiencing a live concert. [Yakura and Goto 2020] examined computational approaches to enhance the sense of co-presence in a VR concert by presenting the movements of virtual audience avatars. [Yan et al. 2020] proposed three interaction modes that arouse social presence for the audience in a virtual performance.

In addition to VR concerts, the intersection between music and Extended Reality(XR) has grown significantly, enabling performers and audiences to interact musically with virtual objects, agents, and environments[Turchet et al. 2021]. Thus, we present Actualities, a live XR performance system that provides audiences with both physical and virtual live performance experience through multiple displays (Figure 1a). We aim to bridge the gap between onsite and online concerts, and aim to explore the potential for interactive engagement within the system.

2 DESIGN AND IMPLEMENTATION

Our system enables onsite audiences to enter virtual concerts and interact with digital elements via various devices. We employ several sensors to capture signals in the performance environment. To capture the guitar signals, we use an arrayed optical pickup that digitizes the vibration signals from the strings. As the guitarist strums the guitar, flowers and butterflies are generated in real-time. For the cajon drum, we use capacitive sensors to digitize the striking signals. As the drummer strikes the cajon, particle explosions appear simultaneously. In addition, millimeter-wave radar sensors are utilized to detect the movement of onsite audiences and transform them into virtual animal avatars. All real-time generative visuals are produced within a 3D forest environment constructed in Unity and projected onto an semi-immersive screen during the concert. The onsite performance is simultaneously live-streamed online, which enable audiences to experience from various viewpoints with different devices. Online viewers can also use reaction buttons on the live-streaming platform to create emoji feedback, and the emoji will be displayed on the onsite screen.

3 EXPLORATORY AUDIENCE STUDY

To explore the user interaction with our system, we recruited 22 participants (3 males, 19 females, aged 23 to 32) for an exploratory study, 10 of whom had attended VR concerts before. We set up a stage and demoed three songs with this system, organizing a mini live performance for a cross-reality experience. Prior to the performance, we explained how the system works and the experimental procedure, providing a QR code for accessing the VR concert. Tablets and VR-HMDs were provided in the experiment, and participants were informed that they can freely watch the performance through different devices. Once participants understood the experimental process, they had the option to create virtual avatars by moving their bodies, send emojis through the live streaming platform, or simply watch the show silently during the performance. Following the experience, participants were asked to complete questionnaires using a 7-point Likert scale and participate in an interview.

4 DISCUSSION AND FUTURE WORK

During the performance, more audiences focused on interactive visual elements such as real-time music visualization, virtual avatars of the onsite audiences, and emoji feedback sent through mobile devices. Participants danced vigorously to create virtual representations (Figure 1b), which was the most active behavior during the performance. However, some participants found it challenging to locate their avatars in the virtual world, which distracted from the performance and diminished their presence. Although most of the participants sent emoji feedback from their mobile devices (Figure 1c), some of them thought the emojis were disruptive and disrespectful to the performers.

Half of the audience watched the performance with VR-HMDs (Figure 1d), while the other half used phones/tablets. Therefore, we performed one-way ANOVA on the questionnaire results to compare the two types of users in terms of presence, enjoyment, engagement, and immersion (Figure 2). The VR-HMD audience had lower scores in all four items, with a significant difference in "presence." According to the interviews, the most frequently mentioned

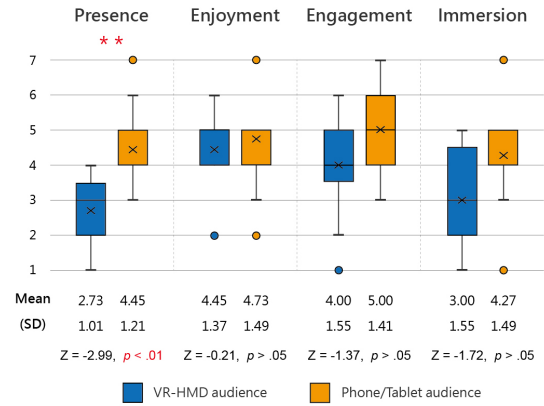


Figure 2: The statistical analysis result of the questionnaire.

negative aspect associated with VR-HMD was the lack of immersion and the feeling of isolation within the virtual world. Conversely, the primary positive aspects highlighted were the "interactive design," including emojis, real-time animal avatars, and music visualization effects, which were reported to be less pronounced in VR.

We offered a live XR performance using multiple displays, allowing audiences to freely choose the reality they wanted to experience. We believed that if they were drawn into the virtual world, they would be interested in trying VR-HMD to enter the VR concert for a better immersion. However, during the transition between realities, the audiences lost their sense of presence and immersion. Some even felt isolated after entering the virtual world when they wore the VR-HMD. To address this issue, we believe that AR could be a solution for those who wish to experience the live performance in different realities while staying connected to the onsite environment. In the future, we plan to incorporate AR into our system and conduct further audience research.

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